Subject:database management system

Total marks: 60

Part A [5 x 2 = 10 Marks]

1.1. What is the primary function of a relational database?

The primary function is to store and manage data in structured tables (relations) that can be linked—or related—based on data common to each, ensuring data integrity and minimizing redundancy.

1.2. Define 'Primary Key' in the context of relational databases.

A **Primary Key** is a specific column (or set of columns) that uniquely identifies each row in a table. It must contain unique values and cannot contain NULL values.

1.3. How does the Entity Relationship Model represent data entities?

The ER Model represents data entities as **Rectangles** (Entity Sets), their attributes as **Ellipses**, and the relationships between them as **Diamonds**.

1.4. What is the significance of 'Foreign Keys' in database design?

Foreign Keys establish and enforce a link between data in two tables to maintain **referential integrity**. They ensure that the value in one table matches a value in the primary key of another table.

1.5. Identify one of Codd's Rules and its purpose.

- **Rule 1: The Information Rule.**
- **Purpose:** All information in a relational database must be represented in only one way, namely as values in a table row/column format.

Part B [2 x 10 = 20 Marks](either a or b)

1.6. a. Evolution of Database Systems

- **Flat-file Systems:** Early data storage where data was kept in separate files. This led to high redundancy, data inconsistency, and difficulty in accessing related data.
- **Hierarchical/Network Models:** Introduced parent-child relationships but were rigid and complex to navigate.
- **Relational Databases (RDBMS):** Introduced by E.F. Codd; uses tables. Key advancements include data independence (logical/physical), SQL for standardized querying, and ACID properties for reliability.

1.6. b. Three-Level Schema Architecture

1. **External Level (View Level):** Describes how users see the data (individual user views).
2. **Conceptual Level (Logical Level):** Describes what data is stored and the relationships between them. It hides physical storage details.
3. **Internal Level (Physical Level):** Describes how the data is actually stored on the storage media (indexes, hashing, etc.).

1.7. a. ER Diagram for Car Insurance Company

- **Entities:** Customer, Car, Accident.
- **Relationships:**
  - **Owns:** Customer (1) to Car (N) — One customer can own many cars.
  - **Involved In:** Car (1) to Accident (N) — One car can be in zero or many accidents.

1.7. b. Relational Algebra Expressions

- i. Name of all red parts: $\Pi_{Pname}(\sigma_{colour='Red'}(Parts))$.
- **ii. Prices for parts that are red or green:** $\Pi_{Price}(Catalog \bowtie (\sigma_{colour='Red' \lor colour='Green'}(Parts)))$.
- iii. SIDs of suppliers who supply a part that is Red/Green: $\Pi_{SID}(Catalog \bowtie (\sigma_{colour='Red' \lor colour='Green'}(Parts)))$.
- iv. Name of all suppliers who supply a part that is red or green: $\Pi_{sname}(Suppliers \bowtie (Catalog \bowtie (\sigma_{colour='Red' \lor colour='Green'}(Parts))))$.

Part A [5 x 2 = 10 Marks]

2.1. What does the SQL 'GROUP BY' clause achieve?

It groups rows that have the same values into summary rows, typically used with aggregate functions (COUNT, MAX, MIN, SUM, AVG).

2.2. Give an example of a SQL data definition command.

CREATE TABLE Students (ID INT, Name VARCHAR(50)); or ALTER TABLE, DROP TABLE.

2.3. What is the role of 'Procedures' in PL/SQL?

Procedures are subprograms used to perform a specific action. They encourage code reusability and allow complex business logic to be stored and executed on the server side.

2.4. How do 'Cursors' enhance data manipulation in PL/SQL?

Cursors allow a program to handle a result set one row at a time. This is essential when a query returns multiple rows that need individual processing.

2.5. What is meant by 'Embedded SQL'?

Embedded SQL refers to the practice of placing SQL statements directly inside a host programming language (like C, C++, or Java).

Part B [2 x 10 = 20 Marks] (either a or b)

2.6. a. SQL Query Processing Steps

1. **Parsing:** Checking syntax and semantics.
2. **Optimization:** The **Query Optimizer** generates the most efficient **Execution Plan** based on statistics and indexes.
3. **Execution:** The engine runs the plan and retrieves data.

2.6. b. Discuss the use and benefits of PL/SQL over standard SQL

PL/SQL (Procedural Language/Structured Query Language) is Google/Oracle's procedural extension to SQL. While standard SQL is declarative (telling the database *what* to do), PL/SQL allows you to tell the database *how* to do it using procedural logic.

Key Benefits of PL/SQL

- **Reduced Network Traffic:** Instead of sending multiple SQL queries one by one from the application to the server, a whole block of PL/SQL code is sent at once, significantly reducing communication overhead.
- **Procedural Capability:** Unlike standard SQL, PL/SQL supports loops (FOR, WHILE), conditional statements (IF-THEN-ELSE), and error handling (EXCEPTION), allowing for complex logic.
- **Better Performance:** Since the code is executed directly on the database server, data processing is much faster as it avoids moving large amounts of data to the client application.
- **Modular Development:** PL/SQL allows you to group related logic into reusable units like Functions, Procedures, and Packages.

2.7. a. SQL Queries (Employee/Department Table)

- **i. Detail of employees in "CSE":** SELECT * FROM Employee WHERE department_number = 'CSE';

- **ii. List Eno, Ename, DeptNo, and Dept Name:** SELECT E.eno, E.ename, E.department_number, D.name FROM Employee E JOIN Department D ON E.department_number = D.department_number;
- **iii. Sort Ename descending:** SELECT eno, ename, gender FROM Employee ORDER BY ename DESC;
- **iv. Earns less than average pay:** SELECT * FROM Employee WHERE basic_pay < (SELECT AVG(basic_pay) FROM Employee);
- **v. Dept Name and Count:** SELECT D.name, COUNT(E.eno) FROM Employee E JOIN Department D ON E.department_number = D.department_number GROUP BY D.name;

2.7. b. Cursors in PL/SQL

Cursors act as pointers to the context area.

- **Steps:** DECLARE -> OPEN -> FETCH -> CLOSE.
- **Importance:** They are vital for transaction processing where row-by-row validation is required before committing changes.